

Team 8 Executive Summary

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Problem

Lackey Clinic is a free and low-cost health clinic in Yorktown, Virginia, whose goal is to treat and perform preventative care for patients to keep them out of hospitals and acute care centers. One of the key metrics that Lackey focuses on is patient churn. Churn is a problem for most businesses, and it especially concerns Lackey because patients need to come in regularly to be properly treated and prevent worsening outcomes. Furthermore, it is expensive and time-consuming for Lackey to onboard patients due to the complex regulatory landscape surrounding healthcare. Our team sought to classify which patients were most likely to churn using statistical modeling and machine learning techniques. We presented our findings to clinic representatives and other stakeholders in the form of an elevator pitch, posterboard, and the following written report. Here we discuss context, methodology, findings, recommendations, and limitations.

Methodology

Given raw data on patient demographics, diagnoses, and visits, we created a custom churn definition and trained two machine learning models with two goals in mind—inference and prediction. To understand the motivation of many modeling decisions, it is first important to understand how we characterize churn. Our definition of churn states that a patient has churned if either of the following is true: the patient has not had an encounter in the past year, or if their current enrollment status is “disenrolled.” We justified this definition by reasoning that one of Lackey’s primary goals as a clinic is to provide regular, preventative care to its patients, and that even someone in perfect health is recommended to visit the doctor at least once per year.

We cleaned our data, joining all data files together to create patient-level summaries with

a single patient and all of their associated information in each row, then performed feature engineering to generate some new predictors. For instance, we used the zipcodeR package to calculate the distance from the center of the patient's zip code to Lackey clinic's coordinates, with the idea that patients further away might be more likely to disengage.

Our first model uses logistic regression to calculate the probability that a given person will churn. This model was primarily selected for its greater explanatory power compared to our other technique, and we used cross-validation to verify its performance. Using logistic regression, we are able to determine the quantitative effect and significance level that every predictive factor has on the outcome variable churn.

The second model we selected has a higher predictive power than logistic regression, being less explanatory and more suited to prediction. This model is KNN, or K Nearest Neighbors. Using similar predictors as logistic regression, we cross-validated our model and achieved an accuracy of 65% and a sensitivity of 86%.

Findings

Based on the output of our logistic regression model—which had lower performance but higher interpretability than KNN—we found that the following variables were significant predictors of churn.

- I. **Employment:** Patients have a higher churn rate if they are Self Employed or Unemployed (Actively Looking For Work). These patients may have less predictable work schedules, or competing responsibilities that make it difficult to follow up on visits.
- II. **Chronic disease:** Patients have a higher churn risk if they do not have a chronic disease on record (i.e., use Lackey's resources only for acute care needs). We hypothesize that

this is because patients with chronic conditions come into Lackey regularly for care, whereas patients with acute conditions may view care on a more one-off basis.

- III. **Housing:** The “lives with others” group had the highest churn rate, at approximately 75%, and the second-highest churn rate was the homeless group. Those who own and rent had similar churn rates. Patients with less stable housing conditions may have to move away from Lackey or not have reliable transportation.
- IV. **Language:** Patients have a lower churn risk if they are not native English speakers. We hypothesize this is because Lackey excels at multilingual support, and patients have a difficult time finding care that meets their language needs elsewhere.
- V. **Age:** Older populations have a higher churn risk. Over 75% of patients ages 65+ will churn, and we assume this high rate is due to patients aging into Medicaid.

In terms of prediction, we discussed earlier how our KNN model was superior at identifying churners before they left Lackey. This model was built to predict whether a new patient will churn, rather than explain in general terms what types of patients are churning. KNN was our most accurate model and with it, we correctly predicted 65% of patients' churn status, which is a 9% increase over the baseline 56%. This baseline is calculated by simply predicting a retention for every patient. Our sensitivity of 86% indicates that of every 100 people who churn, the model correctly flags about 86 of them and misses only 14. Sensitivity is the key metric that we care about, and we tried to get it as high as possible—even at the cost of specificity. This is because Lackey loses little from identifying someone as a churning who is actually not, but they stand to lose much more from failing to catch someone who actually will churn.

Recommendations

Given this context about the patient churn predictors, we recommend the following:

1. Lackey should suggest telehealth follow-ups as a more convenient option for patients who are self-employed or looking for work.
2. Lackey clinicians should emphasize the importance of regular check-ins and preventative care to patients who come in for acute conditions.
3. The high percentage of patients churning due to Medicaid eligibility is notable, but there is nothing Lackey needs to change in this regard—if anything, we hope it comes as a relief to know that 17.72% of churn is accounted for by this population.
4. Lackey should put homeless patients or patients living with others in touch with support services to encourage them to find more stable housing conditions, as this may help with the ability to follow through with care.
5. Lackey should continue to provide strong language support services to patients who do not speak English as their primary language, as these patients likely struggle to find other healthcare providers that support their language needs.

Limitations

A major limitation of this study is a lack of qualitative input from patients—we draw our assumptions from data, but patients are real people. Also, there may be additional nuances we could have accounted for when defining “churn,” such as the recency of a patient’s enrollment, because patients who joined very recently may be classified as churners under our definition. Despite this, we have provided Lackey with interpretable and actionable results, while still positively predicting 84 out of 100 churners.